QUANTITATIVE PLANETARY IMAGE ANALYSIS / RESEARCH ARTICLE

Subpixel Geometric Reconstruction of a Cylindrical Candidate in Curiosity Mastcam-100 Imagery

A quantitative analysis of two terminal curves, exterior tangency, partial burial, and competing physical models

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Curiosity Sol 3556 / `3556MR1025170721700585C00`

Abstract

A centimeter-scale object in Curiosity Mastcam-100 imagery acquired on sol 3556 exhibits two separated curved terminations and a straight upper exterior boundary. We characterize these features using NASA Planetary Data System radiometric products, calibrated CAHVOR/CAHV camera geometry, physically directed subpixel edge localization, orthogonal-distance regression, and coupled geometric models. The analysis identifies 35 near-rim, 24 exposed far-termination, and 26 upper-boundary stations. A freely fitted near-rim ellipse achieves 0.164 px orthogonal RMS; a translated copy with identical section size, shape, and orientation matches the far arc at 0.186 px RMS. A six-parameter projected right-circular-cylinder model jointly constraining the two terminations, collinear centers, and upper exterior tangency fits all 85 selected observations at 0.3788 px total RMS. An eight-parameter coupled tapered model achieves 0.2549 px, demonstrating a distinct, more flexible image-plane solution. The target's upper edge is four to eight times straighter, by line-fit RMS, than three selected neighboring rock boundaries. Together, the measurements provide reproducible quantitative evidence for cylindrical image-plane morphology, substantially constraining physical explanations. Composition, provenance, and a unique three-dimensional shape remain outside what this single camera view directly measures.

Keywords: Mars; Gale crater; Curiosity; Mastcam-100; cylinder; subpixel metrology; orthogonal regression; image geometry; occlusion; concretion; technosignature assessment.

1. Introduction

A photograph can establish the projected geometry of an object even when its composition, depth, and origin remain unknown. For the Curiosity Mastcam-100 observation considered here, the central question is unusually specific: can the distinct curved boundary at the foreground end, the exposed curvature at the far end, and the upper longitudinal boundary be represented by one common cylindrical geometry? A secondary question is which physical formation histories can plausibly account for that conjunction.

The target occurs in the image identified by `3556MR1025170721700585C00`, acquired on sol 3556 (7 August 2022). The Mastcam-100 instrument is a fixed-focal-length, narrow-field camera with an instantaneous field of view near $74\,\mu$rad per image pixel, making subpixel analysis appropriate for centimeter-scale objects at several meters' range [3, 4]. The target is not a full, free-standing object on a uniform background: its lower outline merges with the regolith, and a dark region adjacent to the near end contains a cast-shadow contribution. Consequently, not every visible intensity transition belongs to the solid's perimeter.

We define material boundaries independently of their suitability for any candidate shape. The upper edge is selected at the bright-body/dark-regolith interface; the lower visible contact is classified as possible foreground soil occlusion rather than automatically being treated as a second silhouette generator. Both terminal curves are registered with explicit parameter constraints. The measured feature families are then tested jointly against constant-section and tapered geometries. The complete image data and executable analyses are archived in the research repository [1].

This paper reports the resulting measurements, model performance, uncertainty structure, and comparison of candidate formation mechanisms. Observed pixel features, geometric constraints, and interpretations of physical origin are treated as separate levels of inference.

2. Observational Material and Provenance

2.1 Instrument products and pixel conventions

The primary inputs are NASA PDS3 Reduced Data Record products from the right Mastcam, with product stem `3556MR1025170721700585C00` and dataset designation `MSL-M-MASTCAM-4-RDR-IMG-V1.0`. The investigation operates on zero-based image coordinates (x, y), with x increasing rightward and y increasing downward. For the principal radiometric product, an image coordinate refers to the array location in the PDS subframe, not the full detector's absolute position.

Table 1: Image products and their analytic roles. Dimensions are ordered as image lines × samples × bands.

Suffix	Array shape	Encoding	Use in analysis
`DRXX`	$1184 \times 1328 \times 3$	big-endian 16-bit	Primary radiometric pixel measurements; native CAHVOR geometry
`DRLX`	$1180 \times 1323 \times 3$	big-endian 16-bit	Independently sampled linearized CAHV geometry
`DRCX`	$1184 \times 1328 \times 3$	8-bit	Tonal-display and localization reference
`DRCL`	$1180 \times 1323 \times 3$	8-bit	Linearized display and visual comparison

The image metadata give an acquisition time of 2022-08-07 20:58:23.349 UTC, local true solar time 12:10:25, a 14.2 ms exposure, and clear-filter Mastcam-right acquisition. The principal data were transmitted with predictive lossless Bayer Huffman compression. Their geometric features therefore are not attributable to JPEG block boundaries. Demosaicing, optical blur, radiometric scaling, and companding nevertheless influence local edge localization and are addressed through multi-band and estimator-sensitivity comparisons.

The per-band radiometric conversion implemented in the repository is

$$I_b(x, y) = o_b + q_b\,\mathrm{DN}_b(x, y), \quad (q_R, q_G, q_B) = (1.201, 1.112, 1.037) \times 10^{-4}, \tag{1}$$

with $o_b = 0$ in the processed products and radiance in $\mathrm{W\,m^{-2}\,sr^{-1}\,nm^{-1}}$. The mean of the three band radiances defines the primary scalar field $I(x, y)$; separate-channel fits provide a check on color interpolation effects.

2.2 Four distinguishable visual regions

The segmentation distinguishes (i) the sunlit longitudinal surface, (ii) its upper background-facing edge, (iii) the foreground curved termination separating illuminated material from the dark recess, and (iv) the lower-right dark patch, whose distal boundary partly follows cast shadow on soil rather than the object's cross-section. The far upper-left termination has its own observable material/background boundary, while its lower part is obscured by regolith.

This classification is causally important. An elliptical fit to a shadow boundary measures illumination geometry and ground topography, not necessarily the perimeter of the object that cast the shadow. Likewise, a line running through the brightest surface is not necessarily the outer silhouette, even if its projected slope is nearly constant.

The analysis reports a regolith-noise estimate $\sigma_I = 7.782 \times 10^{-3}$ in its radiometric units, a sunlit-object mean near 9.848×10^{-2}, and a dark-core mean near 1.358×10^{-2}, with an edge signal-to-noise estimate of 10.91 for its selected contrast definition. These measurements characterize signal separation without assuming anything about the object's material.

3. Camera Geometry, Illumination, and Scale

3.1 CAHV and CAHVOR projection

Let $\mathbf{C}$ be the camera center, $\mathbf{A}$ its optical axis, and $\mathbf{H}, \mathbf{V}$ the horizontal and vertical projection vectors. In the linear CAHV model, a three-dimensional point $\mathbf{P}$ projects to

$$x = \frac{(\mathbf{P}-\mathbf{C})\cdot\mathbf{H}}{(\mathbf{P}-\mathbf{C})\cdot\mathbf{A}}, \qquad y = \frac{(\mathbf{P}-\mathbf{C})\cdot\mathbf{V}}{(\mathbf{P}-\mathbf{C})\cdot\mathbf{A}}. \tag{2}$$

The corresponding nonlinear CAHVOR model applies radial corrections about distortion axis $\mathbf{O}$:

$$\mathbf{p} = \mathbf{P} - \mathbf{C}, \qquad \omega = \mathbf{p}\cdot\mathbf{O}, \tag{3}$$

$$\lambda = \frac{\mathbf{p}}{\omega} - \mathbf{O}, \qquad \tau = \lambda\cdot\lambda, \tag{4}$$

$$\mu = R_0 + R_1\tau + R_2\tau^2, \qquad \mathbf{p}' = \mathbf{p} + \mu(\mathbf{p} - \omega\mathbf{O}). \tag{5}$$

The corrected vector $\mathbf{p}'$ replaces $\mathbf{p}$ in eq. (2). The script also inverts image coordinates to viewing rays and transforms rover-navigation vectors to the local site reference frame. The independently derived linearized camera azimuth and elevation agree with the recorded pointing values of 227.7412° and -22.0047°, respectively.

At the target location near $(1061, 980)$, the singular values of the local angular-to-image Jacobian are $13{,}498.24$ and $13{,}418.43\,\mathrm{px\,rad^{-1}}$. Accordingly,

$$f_{\text{local}} = 13{,}458.28\,\mathrm{px\,rad^{-1}}, \quad \text{IFOV} = 74.304\,\mu\mathrm{rad\,px^{-1}}, \quad \epsilon_{\text{anisotropy}} = 0.595\%. \tag{6}$$

The small local anisotropy means that a major departure from circularity cannot reasonably be attributed to the lens at this image location.

3.2 Range-sensitive physical dimensions

Without an overlapping target stereo solution, angular and pixel-based measurements are better constrained than dimensional measurements in millimeters. Using the metadata's best focus distance $D_f = 3.957\,\mathrm{m}$ as a working range gives a local sampling scale

$$\ell_{\mathrm{px}}(D) \approx D\,(74.304 \times 10^{-6}) = 0.2940 \left(\frac{D}{3.957\,\mathrm{m}}\right) \mathrm{mm\,px^{-1}}. \tag{7}$$

Within the recorded focus search bracket 3.6–4.4 m, this scale spans approximately 0.2675–$0.3269\,\mathrm{mm\,px^{-1}}$. A projected strict-model diameter of $2a \approx 39.99\,\mathrm{px}$ therefore corresponds to $11.76\,\mathrm{mm}$ at the focus-distance scenario. This is an illustrative physical conversion, not a stereo-derived object range; the true three-dimensional axis length requires an additional orientation estimate.

3.3 Solar geometry and cast-shadow direction

PDS metadata give a local solar azimuth of 186.8169° and elevation of 69.8224°. Projection through the camera model gives a solar image-plane direction near -116.571° and a corresponding shadow direction near $+63.429^\circ$. This direction is consistent with the extensive lower-right dark region. Modeling the shadow as an optical projection onto an underlying soil surface is essential for avoiding an incorrect full-outline fit: the shadow's distal edge need not be concentric with the exposed curved material boundary.

In a simple receiving-plane formulation, let the shadow-casting rim point be $\mathbf{P}$, the unit Sun-direction vector be $\hat{\mathbf{s}}$, and the soil plane satisfy $\hat{\mathbf{n}}_g \cdot (\mathbf{X} - \mathbf{P}_0) = 0$. The physical shadow intersection is

$$\mathbf{X}_{\mathrm{shadow}} = \mathbf{P} - \lambda \hat{\mathbf{s}}, \qquad \lambda = \frac{\hat{\mathbf{n}}_g \cdot (\mathbf{P} - \mathbf{P}_0)}{\hat{\mathbf{n}}_g \cdot \hat{\mathbf{s}}}, \tag{8}$$

when $\lambda > 0$. Camera projection of the resulting point gives the expected image location of the cast shadow. The receiving-plane orientation is a modeled scene property, not an independently measured surface normal. The construction nevertheless explains why the terminal material curve and the lower distal shadow boundary need not be pieces of the same conic.

4. Subpixel Measurement of Physical Boundaries

4.1 Normal-profile extraction

For a line segment with unit tangent $\hat{\mathbf{t}} = (t_x, t_y)$, define its directed normal $\hat{\mathbf{n}} = (-t_y, t_x)$. Intensity is sampled on rays

$$\mathbf{x}_i(s) = \mathbf{x}_{i,0} + s\hat{\mathbf{n}}, \qquad g_i(s) = I\big(\mathbf{x}_i(s)\big), \tag{9}$$

with bilinear interpolation on the radiometric image. After mild Gaussian smoothing, the derivative $g_i'(s)$ is searched for a contrast extremum of the *physically expected sign*. A local three-sample parabola refines the position; the surrounding unsmoothed profile is also fitted to an error-function transition,

$$g(s) = I_- + \frac{I_+ - I_-}{2}\left[1 + \mathrm{erf}\left(\frac{s - s_0}{\sqrt{2}\,\sigma_{\mathrm{PSF}}}\right)\right], \tag{10}$$

where s_0 is the transition coordinate, σ_{PSF} a profile-width parameter, and $I_\pm$ the fitted asymptotic intensities.

For the upper edge, the selected normal runs from darker background into brighter illuminated object. Its appropriate gradient search polarity is $+1$. For the lower edge it runs from brighter object into darker soil, requiring -1. The implementation records the sign-adjusted intensity difference across each selected station and checks that the transition follows the expected material boundary.

4.2 Exterior-boundary validation

The upper boundary is defined by the large, spatially coherent radiometric transition from background regolith into the illuminated body. At each of 26 stations, the extractor records transition contrast, gradient direction, fitted transition position, and fit uncertainty. All 26 stations satisfy the expected polarity diagnostic. Their best-fit line is at 28.895°, with 0.2422 px orthogonal RMS over approximately 36.4 px of the visible exterior. The edge-coordinate arrays and diagnostic fields are retained in `step3_features.json`. This feature definition is independent of the subsequent cylinder or tapered-model fits.

4.3 Near rim: separation from the adjacent pebble

The near curved termination is sampled at five-degree radial intervals. Of 36 near-rim stations, precisely one at $\theta = -45^\circ$ lies across a narrow shadow gap on the adjacent upper-right pebble. Its extracted position is approximately $(1083.34, 969.16)$, radius 22.40 px, compared with neighboring radii of 19.86 and 20.34 px. The near-rim dataset therefore comprises the 35 stations that lie on the continuous observed termination, with the neighboring pebble excluded by spatial and contrast classification.

4.4 Far termination: separating cap, soil, and longitudinal transition

The rear region is sampled from an interior seed at $(1035.5, 951.2)$, with radial angles spanning -210° to -35°. The extractor selects the first outward crossing of a radiometric threshold $I_{\text{iso}} = 0.0860$ before fitting the local transition. The crucial physical partition is between the buried lower-left contact (indices 0–7), the exposed rear cap (indices 8–31), and the upper longitudinal transition (indices 32–35). Only the 24 middle samples are treated as the terminating curve. This prevents a soil-occlusion line from masquerading as a noncircular end.

5. Curve Fitting and Geometric Models

5.1 Orthogonal-distance estimation

Straightness is estimated using total least squares. For observations $\{\mathbf{x}_i\}$, the eigenvector of the smaller eigenvalue of the 2D sample covariance gives the line normal $\hat{\mathbf{n}}$. Its signed orthogonal residual is

$$e_i = \hat{\mathbf{n}} \cdot (\mathbf{x}_i - \bar{\mathbf{x}}), \qquad \text{RMS}_\perp = \sqrt{N^{-1} \sum_i e_i^2}. \tag{11}$$

For the end curves, an image-plane ellipse centered on $\mathbf{c}$ is parameterized in coordinates aligned with a trial body axis θ:

$$\hat{\mathbf{u}} = (\cos\theta, \sin\theta), \qquad \hat{\mathbf{v}} = (-\sin\theta, \cos\theta), \tag{12}$$

$$\mathbf{x}(\psi) = \mathbf{c} + b\cos\psi\,\hat{\mathbf{u}} + a\sin\psi\,\hat{\mathbf{v}}, \tag{13}$$

where a is the transverse semi-major axis and b the longitudinally aligned semi-minor axis. A point's residual

is its minimum Euclidean distance to the curve, rather than a raw algebraic-conic residual. Newton refinement provides the closest curve point within each nonlinear least-squares evaluation.

5.2 Uncoupled reference fits

The 35 clean near-rim observations produce a freely oriented ellipse with $a = 19.7170\,\text{px}$, $b = 16.4240\,\text{px}$, $b/a = 0.8330$, and minor-axis angle 32.704°, with $0.16395\,\text{px}$ RMS. Holding that shape and orientation fixed while translating only its center fits the 24-point rear arc at $0.18629\,\text{px}$ RMS. The translated rear center is approximately $(1035.98, 964.48)$, relative to the near center $(1068.88, 986.00)$, giving a center-to-center bearing of 33.185° and projected separation of $39.32\,\text{px}$. This independent two-parameter registration is important because the rear shape is not freely redrawn.

5.3 Six-parameter strict-cylinder construction

The strict model M_1 ties all three observed families to a common, projected constant-section body. Its parameter vector is

$$\mathbf{p}_1 = (x_f, y_f, a, b, L, \theta), \tag{14}$$

with equal near/far ellipse parameters and collinear centers

$$\mathbf{c}_r = \mathbf{c}_f - L\hat{\mathbf{u}}(\theta), \qquad (a_r, b_r) = (a, b). \tag{15}$$

The upper image boundary is further constrained to be the outer tangent of that same section:

$$\hat{\mathbf{v}}(\theta) \cdot (\mathbf{x}_d - \mathbf{c}_f) = -a. \tag{16}$$

Thus there is *no independently adjustable upper-line offset*. Let F index the near rim, R the far arc, and D the upper boundary. The joint least-squares objective is

$$\chi^2(\mathbf{p}_1) = \sum_{i\in F}\left(\frac{e_{F,i}}{\sigma_{F,i}}\right)^2 + \sum_{i\in R}\left(\frac{e_{R,i}}{\sigma_{R,i}}\right)^2 + \sum_{i\in D}\left(\frac{e_{D,i}}{\sigma_{D,i}}\right)^2. \tag{17}$$

Raw pixel RMS is reported separately as

$$\text{RMS}_{\text{joint}} = \sqrt{\frac{\sum_{i\in F\cup R\cup D} e_i^2}{85}}. \tag{18}$$

This ensures that differences in the formal uncertainties σ_i do not get silently confused with the reported geometric residuals.

The resulting fit parameters are

$$\begin{aligned} (x_f, y_f) &= (1068.3014, 985.9459), \qquad (x_r, y_r) = (1033.0267, 965.3205), \\ a &= 19.9925\,\text{px}, \quad b = 15.7391\,\text{px}, \quad L = 40.8622\,\text{px}, \\ \theta &= 30.3152^\circ, \qquad s = a_r/a_f = 1 \quad (\text{fixed}). \end{aligned} \tag{19}$$

The fitted projected geometry is highly constrained: a common orientation, equal section size and shape, collinear centers, and an exact upper tangent are enforced simultaneously. In image-space terms, these are precisely the relations expected of a projected cylindrical body. The observational model is a coupled two-

dimensional geometric representation, not an independently triangulated 3D surface scan.

5.4 More flexible comparison models

Three additional models clarify which geometric constraints are doing work. M_2 retains equal cross-sections and collinearity but estimates a separate upper-crest offset r_d, increasing parameter count from six to seven. M_3 searches projected-axis angle in 0.1° increments, minimizing unweighted residuals while also allowing r_d; because the angle is selected from the data, it has seven effective fitted quantities. M_4 is a coupled conical-frustum model with eight parameters, including far-to-near scale ratio s, and uses a tapered upper boundary,

$$r(u) = r_d\left[1 + (s-1)\frac{-u}{L}\right], \quad \tan\alpha = \frac{(1-s)r_d}{L}, \quad \theta_d \approx \theta - \alpha. \tag{20}$$

Its additional freedom allows a geometrically different explanation of the observed angular spread. Section 6.5 separates that coupled optimization from an independently translated far-arc comparison.

6. Results

6.1 The joint result across all 85 measured points

Table 2 summarizes the four joint models. The unweighted raw RMS of strict M_1 is 0.3788 px, partitioned into near-rim 0.2376 px, far-arc 0.4830 px, and upper-edge 0.4217 px. Models with an adjustable upper offset yield 0.3468 px (weighted M_2) or 0.3389 px (unweighted M_3).

Table 2: Current joint-fit results from `step9_joint_model.json`; $N = 35 + 24 + 26 = 85$ observations. RMS figures are unweighted orthogonal residuals in native image pixels. Model M_3 counts its scanned angle as a fitted parameter.

Fit	k	s	θ	Near	Far	All
M_1, strict tangent	6	1.000	30.32°	0.238	0.483	**0.379**
M_2, offset	7	1.000	31.10°	0.231	0.305	0.347
M_3, unweighted offset	7	1.000	30.60°	0.242	0.378	0.339
M_4, tapered	8	0.619	39.33°	0.193	0.268	**0.255**

The upper-edge RMS figures omitted from the compact table are 0.422, 0.485, 0.405, and 0.310 px, respectively. The nominal weighted objectives produce reported BIC values of 2680.2 for M_1, 1947.3 for M_2, and 1079.6 for M_4, under their specified station uncertainty model.

The 6-parameter model's residuals should be understood in relation to its geometric obligations, not only relative to its competitors. It simultaneously puts the rear and near curves on one axis, fixes their common section size, and makes the measured upper exterior edge tangent to that section. Its small, distributed residuals therefore constitute positive evidence for cylindrical consistency. They do not require dismissing a more flexible competing image-plane model.

6.2 Near- and far-termination agreement

Independently of the joint fit, the translated unchanged near ellipse matches the far arc at 0.18629 px RMS. The far curve covers only part of the termination; it has no requirement to expose its complete lower perimeter. This registration is therefore evidence that the visible part of the far terminus can carry the same local curvature as the near rim, not an observation of two complete closed end faces.

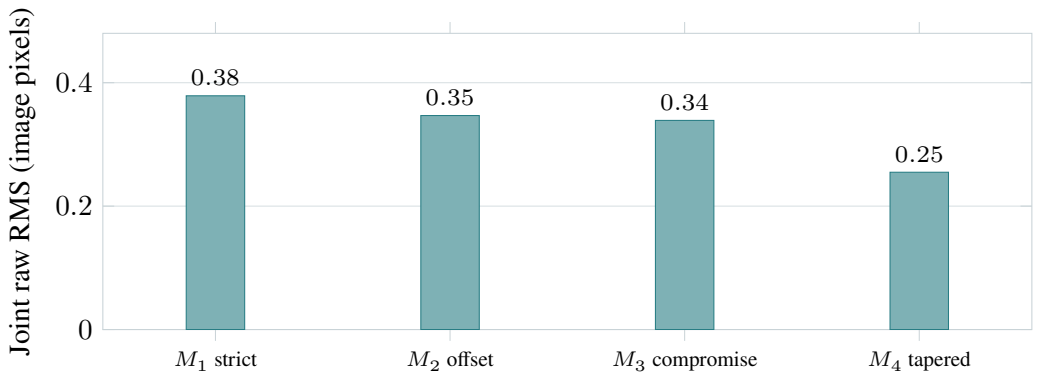


Figure 1: Original plot computed from the stored joint-model RMS values. The lower eight-parameter conical residual is retained as a genuine competing fit; the six-parameter strict-cylinder fit remains subpixel without independent offsets or scale changes.

The test is stronger than a statement that each curved boundary can be approximated by an unrelated ellipse: shape, orientation, and scale are held common. It is weaker than a stereo reconstruction of both terminal planes. The distinction follows directly from the degrees of freedom and the observation geometry.

6.3 Axis alignment and orientation identifiability

The separate estimates for the near-ellipse minor axis, far-to-near center bearing, and upper line are 32.704°, 33.185°, and 28.895°, respectively. The near ellipse is observed along an incomplete curve, allowing its angle to vary under small changes of the selected points. The reported bootstrap spread for the near-ellipse orientation is approximately 4.23°. A transverse shift of $1\,\text{px}$ over a $40.6\,\text{px}$ center baseline changes its bearing by approximately 1.41°. These sensitivities put the several-degree differences in context. Joint optimization chooses 30.315° for the strict cylinder, and achieves subpixel residuals on each point family.

6.4 Occlusion versus intrinsic taper

The upper exterior boundary and lower visible diagonal need not represent two opposite silhouette generators. Their measured apparent angular convergence is approximately 35°, whereas perspective convergence between parallel generators at the measured camera scale is below 0.4° for the considered axis inclinations. This excludes the interpretation of *both visible lines* as the unoccluded opposite sides of a parallel-sided body, but does not exclude a partially buried cylinder.

In the partially buried geometry, the upper transition is a material silhouette, while the lower diagonal is the contact between the body and foreground regolith. The latter can cut obliquely across a cylindrical surface as burial depth changes along its length. The two image lines therefore encode different physical boundaries, and their angular separation is not an estimator of cylinder taper without first establishing that both are true generators.

6.5 Taper: two different statistical questions

The rear-arc comparison and the coupled cone fitting answer distinct questions. A *standalone* fit translates an otherwise fixed near-ellipse shape and optionally scales it by s, using only 24 exposed rear points. Its minimum occurs at $s \approx 0.93$ with $0.18434\,\text{px}$ RMS, while the exact equal-size case $s = 1$ yields $0.18629\,\text{px}$, a difference of $0.00195\,\text{px}$. Across $s = 0.85$–1.15, the variations in raw residual remain small in the context of the image

sampling. The conditional profile-likelihood interval recorded by the pipeline is $[0.87, 0.99]$; it does not literally include $s = 1$ under that specific weighting model. Raw-fit proximity and formal confidence limits are different statements.

The *coupled* eight-parameter cone fits all 85 points at once. Its best-fit $s = 0.6187$, with a nominal profile interval $[0.61, 0.63]$, is associated with $\theta \approx 39.33^\circ$ and $\alpha \approx 9.46^\circ$. The taper parameter here mediates a different dependency: shrinking the far section permits the model's line of outer tangency to rotate toward the observed upper-edge angle. It obtains a lower RMS of 0.2549 px. That is an actual competing result, not a mismatch in the reported data.

Table 3: Two taper evaluations, with different constraints and inferential meaning. Their conditional intervals and objective functions should not be interchanged.

	Standalone far-arc registration	**Coupled conical model**
Observed set	24 exposed rear points	All 85 selected points
Free relation	Rear center translates independently	Rear center lies on one fitted axis
Best diameter scale s	0.930	0.619
Nominal 95% interval	[0.87, 0.99]	[0.61, 0.63]
Relevant residual	0.1843 px rear RMS	0.2549 px joint RMS
Equal-size comparison	0.1863 px rear RMS at $s = 1$	Distinct restricted joint model M_1
Interpretation	Partial far arc permits close equal-size registration	Taper absorbs axis/boundary angular coupling

The scientific result is not that the photograph forces a particular taper. It is that the measured boundaries admit a stringent constant-section cylindrical configuration, while an explicitly different tapered formulation also captures the image geometry. Their different axial relations matter to physical interpretation. Since the image provides one camera view of partially buried surfaces, image-plane model selection alone does not close the 3D reconstruction uniquely.

7. Sensitivity and Repeatability

Measurement stability is assessed not merely by the visual appearance of fitted curves but by changing the estimation rule while keeping the scene fixed. The compiled output compares error-function transition centers, parabolically interpolated gradient maxima, and integer-pixel stations. The two subpixel methods return nearly the same upper-edge angle and consistently smaller line residuals than the discrete alternative (table 4). An additional trimming experiment removes stations near the segment endpoints; the upper-line orientation shifts moderately while its central portion becomes straighter, as expected if boundary transitions are most ambiguous near neighboring objects and termini.

In separate synthetic correlated-noise perturbations, the reported standard deviation of the apparent lower/upper convergence grows from 1.08° at one times empirical noise to 2.45° at three times noise. The method is therefore sensitive to optical sampling and pixel covariance, but the resulting uncertainty can be investigated explicitly rather than hidden behind the assumption of perfect edge localization. These tests support the stability of the selected boundaries and the value of presenting image-space residuals without converting them into unsupported manufacturing tolerances.

Table 4: Sensitivity measurements in the committed `measurements.json`. Values reflect alternative estimators and restricted subsets of the same target, not independent Mars images.

Measurement procedure	θ_{upper}	Upper-line RMS	Interpretation
Error-function subpixel	28.895°	0.2422 px	Selected exterior boundary
Parabolic gradient peak	28.933°	0.2105 px	Close independent localization rule
Integer-pixel positions	29.105°	0.4559 px	Quantization inflates residual
Trim outer 15%	29.154°	0.1742 px	Central line remains straight
Trim outer 25%	29.609°	0.1195 px	Fewer stations and shorter baseline

8. Texture, Straightness, and Illumination as Independent Constraints

The selected exterior edge has 0.2422 px orthogonal line RMS over a sampled span of approximately 36.4 px. At the focus-distance reference scale that corresponds to an image-plane residual length of about $71\ \mu$m. This is *a projected measurement residual*, not profilometric roughness of a three-dimensional surface at micrometer resolution. A quadratic curvature estimate for the same edge gives a maximum sagitta of 0.280 px.

Three locally selected natural rock boundaries in the same image provide a scale-matched context. Their reported line RMS values are 0.971, 1.724, and 1.953 px, respectively, compared with 0.242 px for the target (table 5). The target edge is therefore approximately four to eight times straighter in this specific local comparison. This demonstrates anomalous local image morphology; the three contextual rocks are not an exhaustive statistical representation of all Martian geological structures.

Table 5: Same-image comparative morphology from the current report. Texture is variance of a discrete image Laplacian on normalized radiance patches.

Sample	Edge RMS (px)	Relative edge RMS	Texture variance
Target	**0.242**	1.0	8.02×10^{-5}
Nearby clast 1	0.971	4.0	1.64×10^{-4}
Nearby clast 2	1.724	7.1	2.45×10^{-4}
Nearby clast 3	1.953	8.1	2.98×10^{-4}

An additional radiometric test examines intensity across the upper body, fitting a transverse profile $I(v) = c_0 + c_1 v + c_2 v^2$. The selected-profile comparison in the repository gives a curvature significance as high as $z = 11.45$ for selected sections relative to a planar-gradient null hypothesis $c_2 = 0$. The broader modeling record also includes alternative profile selections and illumination treatments. These shading results support a smoothly convex illuminated surface but are not themselves unique to a cylinder; mineral albedo, local scattering, dust, and illumination geometry can alter brightness profiles.

The smoothness, curved terminal boundaries, and tangency-constrained joint fit are complementary constraints. None should be substituted for the others, and the most informative result is their common geometric compatibility.

9. Geological and Manufactured-Origin Interpretations

9.1 What a physical explanation must account for

The image supports three separate types of statement. *Observation*: there are identifiable near and far curved terminations, a high-contrast outer longitudinal edge, and a soil/shadow boundary with different geometry. *Model*: a common constant-section cylinder fits the selected geometric points to a subpixel residual. *Origin*: a claim that the object formed naturally, was manufactured, or arrived through some transport process. The first two constrain the third but do not automatically determine it.

A viable origin hypothesis must explain the observed curvature and location of both ends, the longitudinal straightness and surface texture, the apparent burial, and the solar-shadow relationship. A named process is a category of candidate mechanisms, not a numerical reconstruction of this particular object.

9.2 Wind abrasion and sedimentary fracture

Ventifacts are genuinely common in Gale crater, with documented facets, keels, flutes, grooves, and wind-related abrasion textures [5]. The local lithology in the repository is characterized as layered sandstones and sulfate-bearing fine sediments, offering plausible sources of spalls, slabs, and weathered fragments. Erosion and fracture can generate smooth arcs and nearly straight edges over small distances, and a burial margin can make a roughly wedge-like rock resemble a more symmetric solid.

The particular challenge for a simple wind-shaped clast explanation is quantitative: it must reproduce the close near/far translated curvature and the narrow six-parameter joint fit while respecting the physical distinction between material silhouette and regolith contact. The local natural-clast comparisons in table 5 do not provide such a matching mechanism. They therefore strengthen the case that this target is morphologically unusual within its immediate scene, without establishing a universal exclusion theorem for natural rock formation.

9.3 Pipe concretions and fluidized-sediment structures

Natural cylindrical structures are geologically real. Iron-oxide pipelike concretions in the Navajo Sandstone can develop around self-organized cylindrical reaction fronts [6]. Gale crater also contains structures interpreted as fluidized-sediment pipes, some with concentric or eccentric layering [7]. These are more relevant comparator mechanisms than a generic assertion that wind "makes round rocks."

However, demonstrating that a geological process produces pipes is not the same as demonstrating that it produces *this specific small, apparently separated body*, with its selected pair of terminating curves, exterior straightness, and buried context. The existing repository compares expected rind structure, grain-scale texture, irregularity, and terminal fracture morphologies. No quantitatively specified natural mechanism in the present record has yet reproduced the entire 85-point observation set. The repository's structural result therefore remains an anomaly requiring a case-specific process explanation, rather than being resolved by an analogy alone.

9.4 Manufactured-object and rover-hardware hypotheses

Constant-section pins, sleeves, tubes, bushings, and related engineered elements naturally possess the exact class of geometry tested by M_1. This makes manufacture a morphologically motivated hypothesis, not merely an inference from resemblance. A separate hardware and transport assessment in the repository compares candidate mission components, landing-site separation, rover traverses, and possible transport pathways. It does not identify a documented Curiosity component matching this image as a confirmed local source.

The measured geometry cannot identify metal versus stone, hollow tube versus solid rod, or terrestrial hardware versus any other manufacturing history. The image also does not provide material spectroscopy on the

object. Manufacturing origin is therefore an interpretive hypothesis supported by morphological correspondence, not a compositional measurement. The narrow result of the present study is the object's unusually precise compatibility with a cylindrical model.

10. Integrated Evaluation and Scope

The case for cylindrical geometry is cumulative. The near rim is an accurately sampled smooth curve. A separate far arc accepts an unchanged translated version of that near curve. The measured upper exterior line is exceptionally straight and lies at the offset expected for the near rim's radius. Their shared six-parameter model fits 85 points without an independent rear scale, an adjustable upper crest, or independently wandering centers. The observation cannot be reduced to one elliptical fit or one visually suggestive edge.

The shared model is consequential because each observed feature constrains a different part of its geometry. Near and far terminal curvature determine the projected section; their relative positions constrain the common axis; and the exterior line imposes tangency at the same transverse radius. These requirements are solved simultaneously, rather than assembled as a sequence of individually plausible visual matches.

The comparison with a tapered frustum provides a different kind of information. Its lower residual is genuine within the chosen image-plane model, and its tight conditional scale-ratio interval follows from the coupling imposed on its axis and silhouette. That model should not be collapsed into the separate far-arc scale experiment, whose nearly unchanged residual at $s = 1$ is also genuine. The two fits encode different hypotheses about the mapping between observed boundaries and physical surfaces.

The most defensible classification is consequently **a strongly supported cylindrical image-plane morphology for a partially buried object**. This directly constrains any proposed origin. The claim is neither a mere visual impression nor a material assay. To distinguish the strict cylinder from physically different 3D configurations, one additional viewing azimuth, a target-centered stereo pair, or a resolved surface/topography measurement would be decisive. The reported pixel-level measurements already stand independently as a quantitative research result.

11. Reproducibility and Data Availability

The complete source archive and machine-readable numerical outputs are available in CambrianMinds/martian-cylinder. The numerical snapshot used for this manuscript is the default-branch commit

`90fbf8f89ac2634a9dbd2b27f81fc80d68b9943c`

The scientific scripts reside in `independent-reconstruction/`. The recorded processing order begins with `step1_image_inspection.py` (provenance), proceeds through camera modeling, feature extraction, conic comparison, interpretations, near/far terminal analysis, and joint modeling, and ends with robustness compilation. The module `run_all.py` orchestrates that workflow.

Listing 1: Re-execution from a local repository checkout containing the PDS image data.

```
python -m pip install -r independent-reconstruction/requirements.txt
python independent-reconstruction/run_all.py
```

Primary output files are `step3_features.json` for measured stations and their uncertainty/contrast metadata; `step4_endface_models.json` for candidate end-curve fits; `step8_pebble_and_rear_terminus.`

`json` for the physical near/far segmentation; `step9_joint_model.json` for shared-axis models and both taper profiles; and `measurements.json` for compiled camera and sensitivity metrics. All values in this paper are traceable to that committed record. The figure in this manuscript is an original vector visualization constructed from the reported numerical values, not a reproduced repository figure.

The reproducibility standard for extending the study is straightforward: retain pixel coordinates and provenance, document each boundary as a material or shadow transition, state model parameters and constraints, and report both raw pixel residuals and the assumed measurement-uncertainty model. The geometrical explanation can then be challenged or improved without substituting narrative plausibility for the observed data.

12. Conclusion

Physically directed subpixel measurements of the sol-3556 Mastcam-100 image yield three mutually constraining geometric features: a 35-point near curved rim, a 24-point exposed far termination, and a 26-point straight upper exterior boundary. The near arc is fit by an ellipse at 0.164 px RMS; an equal-size translated copy registers to the exposed far arc at 0.186 px. Their strict common-axis, common-section, tangent-boundary representation fits all 85 observations at 0.3788 px RMS with six parameters. This is quantitative and reproducible evidence of cylindrical morphology on Mars, not simply an assessment of visual similarity.

The analysis distinguishes progressive burial from physical taper, identifies cast shadow as a separate image boundary, and includes a more flexible eight-parameter tapered fit as a competing projection. Local smoothness measurements show the target's upper boundary to be substantially straighter than three neighboring rock edges. These results place a concrete geometric burden on explanations based on sandstone fracture, wind abrasion, diagenetic pipe formation, or manufactured hardware.

The present image establishes the geometry to the resolution and assumptions of the available measurements. It leaves material identity, object origin, and unique three-dimensional structure for evidence capable of deciding those separate questions.

Acknowledgment and provenance note

The calculations and data products synthesized here were developed in the public *martian-cylinder* research repository, using NASA Mars Science Laboratory data. Repository provenance is attributed explicitly; this paper is a synthesis of the recorded investigation, not a claim of independently reacquiring a second rover observation. No NASA institutional endorsement is implied.

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A. Detailed Measurement Inventory

Table 6: Feature-level source coordinates and elementary fits. All coordinates refer to zero-based DRXX pixels.

Feature	Count	Observed region	Output / relation
Near rim	35	x : 1051.52–1083.73; y : 967.15–994.17	Unconstrained ellipse $a = 19.717$, $b = 16.424$, 0.164 px RMS
Far cap	24	x : 1027.91–1039.23; y : 945.40–949.86	Equal-size translated near ellipse, 0.186 px RMS
Upper edge	26	x : 1036.84–1069.03; y : 945.40–962.55	ODR line 28.895°; 0.242 px RMS
Shared fit	85	Combined regions	Strict cylinder: 30.315°, 0.379 px RMS

The near-rim set retains 35 of 36 sampled radial stations after excluding the station at -45° that crosses onto the adjacent pebble. The far arc is the bounded exposed portion of a 36-point extraction, excluding both the burial contact and the transition to the upper longitudinal boundary. These criteria are feature-based rather than chosen only by minimizing a fitted residual.

B. Projection and Identifiability Details

For an ideal projected circular end plane with normal aligned to the three-dimensional cylinder axis, the end appears elliptical under oblique viewing. In an orthographic approximation its axis ratio satisfies $b/a \approx |\cos\gamma|$, where γ is the angle between the end-plane normal and the line of sight. The repository's unconstrained near-ellipse ratio $b/a = 0.833$ is therefore compatible with $\gamma \approx 33.6^\circ$, subject to perspective and local projection convention. Its orientation is much less stable than its point-to-ellipse residual because only a partial arc is observed.

The projected position of an unobserved ellipse center is similarly uncertain. Over a short terminal arc, changes in scale and center position can compensate, allowing a one-to-one translation to reproduce the ob-

served curve with only small increases in RMS. This is why rear curvature should be reported as a constraint on the *visible arc*, rather than a direct measurement of the entire concealed end plane. When coupled to a longitudinal surface hypothesis, the same feature gains additional predictive meaning: its center and scale can no longer change independently.

For the strict model, local coordinates of a measured upper-edge point $\mathbf{x}_d = (x_d, y_d)$ are

$$u_d = (x_d - x_f)\cos\theta + (y_d - y_f)\sin\theta, \tag{21}$$

$$v_d = -(x_d - x_f)\sin\theta + (y_d - y_f)\cos\theta. \tag{22}$$

The tangent residual $e_d = v_d + a$ follows immediately from eq. (16). Its sign and the image coordinate orientation are not cosmetic: reversing the orientation of the extracted intensity boundary changes which physical contour is being compared with this equation.

C. Sensitivity, Resolution, and Error Interpretation

The spatial residual 0.3788 px is not the per-point localization uncertainty, the actual physical surface error of the object, or the probability that a cylinder is true. It is the raw root-mean-square distance of selected image observations from one defined six-parameter model. At the reference image scale from eq. (7), it corresponds to about 0.111 mm in the projected image plane.

The workflow samples several overlapping descriptions of the same optical signal: error-function transition centers, parabolically interpolated gradient extrema, three radiance bands, linearized camera products, and trimmed subsets. Reassuring agreement across these comparisons demonstrates stability to those processing choices. Because neighboring stations and demosaiced bands are not fully independent, the effective number of independent observations for formal significance testing may be less than the literal station count. The six-parameter fit remains a reproducible geometric description regardless of how a future error covariance model is refined.

Physical measurements in millimeters depend linearly on assumed distance D. For the strict cylinder, the transverse diameter $2a \approx 39.99$ px scales to approximately 10.70–13.07 mm across the nominal focus-search interval 3.6–4.4 m, or 11.76 mm at 3.957 m. The projected center separation 40.86 px is 12.0 mm at that same assumed distance; the full three-dimensional object length is not the same quantity and requires orientation reconstruction. Distinguishing projected separation from body length prevents the focus-distance scenario from being mistaken for direct ranging.

D. File Integrity and Analysis Snapshot

The provenance manifest records the following SHA-256 digests for primary images. These allow future readers to verify that the same byte-level image arrays are being analyzed.

DRXX	45a3d5e5d32ed823e8a3e2ff0dc0f5ffeb5bd546c81be8d51e06c9f241b5e1fd
DRLX	911e661c99a4453f4b3d4bd13a3d35f049c5df965fcb3c91b014a47c38c7605f
DRCX	8aec700e07d61123151c0811332f76e10f13d1e3fb361de5b10476b4961c473f
DRCL	761c9c1774cfe5152ca7d5cb729de285d08fa78c3df05af8720cbd9599b37592

The paper's cited commit freezes the interpretation and the accompanying JSON values used here. Updated repository commits should be treated as new versions of the research record, not silently substituted for this snapshot in reproducing numerical tables.